PROJECT 01 | DATA SCIENCE & AI

Exploratory Data Analysis (EDA) on Titanic

Beginner | About 6 hours | Data Science & AI

WHAT YOU WILL BUILD

Load and analyze the classic Titanic or Student Performance dataset using Pandas to find key statistical insights.

Load the Titanic passenger dataset, clean it, and answer questions such as who was most likely to survive, using pandas.

YOUR REQUIREMENTS

- DataFrame operations (head, info, describe)
- Handling missing values and duplicates
- Grouping and aggregation analysis

WHAT YOU WILL LEARN

- Loading and inspecting a DataFrame
- Finding and fixing missing values and duplicates
- Grouping and aggregating data
- Turning numbers into a written insight

MATERIALS YOU NEED

TOOL	**Python 3.10+ with Jupyter**	Install Python from python.org, then pip install jupyter pandas numpy matplotlib seaborn scikit-learn. Google Colab is a free alternative that needs no install.
DATA	**Titanic dataset**	Load it in one line with seaborn: sns.load_dataset('titanic'). It downloads a small CSV the first time you run it.
READING	**10 minutes to pandas**	The official getting started guide at pandas.pydata.org covers every function you need.
TEMPLATE	**Question list**	Write three questions first, for example: Did women survive more than men? Did ticket class matter? Was age related to survival?

STEP-BY-STEP GUIDE

1 Write your questions

Before touching the data write three questions you want to answer. This keeps your analysis focused.

2 Load and look

Load the dataset and run df.shape, df.head(), df.info() and df.describe(). Write down what each column means.

3 Find missing values

Use df.isna().sum() to see the gaps. Note that age and deck have many missing values.

4 Fix them with a reason

Fill age with the median age of the same class and sex, drop the deck column because it is mostly empty and fill embarked with the most common value. Write why you chose each fix.

5 Check duplicates and types

Count duplicate rows with df.duplicated().sum() and make sure numeric columns are numbers and categories are categories.

6 Group and compare

Use groupby to compute the survival rate by sex, by class and by age band (create bands with pd.cut). Sort the results.

7 Answer your questions

For each question write one clear sentence with a number, for example Women survived at about 74 percent compared with about 19 percent for men.

8 Tidy the notebook

Add headings and notes between code cells, restart and run all cells to prove it works, and export to HTML or PDF.

WORKED EXAMPLE OF A FINISHED PROJECT

Clean and compare

Core steps. Add your own questions and notes.

```
import pandas as pd
import seaborn as sns

df = sns.load_dataset("titanic")
print(df.shape)
print(df.isna().sum())

# Fix missing values and explain why in your notebook
df["age"] = df["age"].fillna(df.groupby(["pclass", "sex"])["age"].transform("median"))
```

```
df["embarked"] = df["embarked"].fillna(df["embarked"].mode()[0])
df = df.drop(columns=["deck"])
print("Duplicate rows:", df.duplicated().sum())

# Compare survival rates
print(df.groupby("sex")["survived"].mean().round(2))
print(df.groupby("pclass")["survived"].agg(["mean", "count"]).round(2))
df["age_band"] = pd.cut(df["age"], [0, 12, 18, 40, 60, 100],
                        labels=["Child", "Teen", "Adult", "Middle-aged", "Senior"])
print(df.groupby("age_band", observed=True)["survived"].mean().round(2))
```

CHECK YOUR WORK BEFORE YOU SUBMIT

- [ ] Three clear questions are written at the start
- [ ] Missing values are handled and the reasons explained
- [ ] Groupings compare survival by at least three factors
- [ ] Each question has a one-sentence answer with a number
- [ ] The notebook runs from top to bottom without errors
- [ ] Headings and notes make it easy to follow

WHAT TO SUBMIT

- A link to your notebook (GitHub or Colab with sharing on) or an exported HTML file
- A short written summary of your three findings

You submit a link. Put your files in Google Drive, GitHub, Figma, YouTube or similar, allow anyone with the link to view them, and paste the link when you submit.

COMMON MISTAKES TO AVOID

- Do not drop rows blindly; explain every cleaning choice.
- Correlation is not cause; word your findings carefully.
- Restart the kernel and run all cells before you submit.
- Use real numbers in your summary, not just words like many.

EXPECTED OUTCOME

Master Pandas library and exploratory data cleaning techniques.

Submit your work from the My Tasks page of your dashboard. A mentor reviews it and sends feedback; if changes are requested you can update and resubmit. Write your own solution - the example above shows the standard to aim for, not something to copy.